

AI-Driven Health Mobile Application for Hypertension and Diabetes Prevention

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Abstract— In Sri Lanka, diabetes and hypertension are regular non-communicable diseases and significant public health issues. Lifestyle variables like poor eating habits, long-term stress, and sedentary activity are closely linked to these illnesses. However, rather than emphasizing early prevention, current healthcare systems mostly concentrate on diagnosis and treatment. An AI-powered mobile health platform is proposed in this study to help Sri Lankans prevent diabetes and hypertension at an early stage. The system offers proactive health monitoring and customized recommendations by integrating many artificial intelligence technologies into a smartphone application. The platform consists of four primary modules: a domain-specific conversational AI agent for stress monitoring, AI-powered food recognition with nutritional analysis, diabetes risk prediction, and hypertension risk prediction.

The system's main innovation is the stress monitoring module. It analyzes user interactions and measures stress levels using metrics like the Perceived Stress Scale (PSS-10), utilizing Natural Language Processing and Retrieval-Augmented Generation. The system offers customized stress management and lifestyle suggestions based on identified stress levels. The proposed platform seeks to encourage preventative healthcare practices and raise early health awareness among Sri Lankans.

Keywords— *Perceived Stress Scale, Retrieval-Augmented Generation, Natural Language Processing, Hypertension, Artificial Intelligence*

I. INTRODUCTION

Among the most important non-communicable diseases affecting people worldwide are diabetes and hypertension. Serious health issues like cardiovascular disease, stroke, kidney failure, and metabolic problems are exacerbated by these circumstances. Due to lifestyle changes like poor eating habits, inactivity, and elevated stress levels, the prevalence of both diseases is rising. Diabetes and hypertension are now serious public health issues in Sri Lanka. Adults who lead sedentary lifestyles, eat poorly, and experience stress at work are far more likely to develop these illnesses. Stress is a significant contributor to hypertension. However, a lot of people don't take preventive health measures or routinely check their stress levels.

Opportunities for early prevention and lifestyle intervention are limited by traditional healthcare systems' primary focus on identifying illnesses after symptoms manifest. Digital health platforms have become useful instruments for proactive health monitoring and disease prevention due to the quick development of artificial intelligence and mobile technology.

In order to avoid diabetes and hypertension in Sri Lankans, this study suggests an AI-driven mobile health platform. The

system offers health monitoring and tailored recommendations by combining many artificial intelligence technologies into a single smartphone application.

The AI stress monitoring agent is the system's primary innovation among these elements. To identify stress levels and offer tailored advice for stress management and hypertension prevention, the agent makes use of conversational AI and validated stress assessment techniques.

This study's primary contributions are as follows:

- Creation of a model for predicting the risk of hypertension using AI
- Using lifestyle data to implement a diabetes risk prediction system
- Creation of an AI-driven module for food identification and dietary advice
- Creation of a conversational AI agent for stress monitoring that is domain-specific

II. LITERATURE REVIEW

In order to promote early disease prediction, individualized health monitoring, and preventive healthcare management, artificial intelligence technologies are being used more and more in the medical field. Conversational AI, computer vision, and machine learning systems have shown great promise in identifying health hazards, examining behavioral patterns, and making health recommendations. This section examines earlier studies on stress monitoring, conversational AI in mental health assistance, food monitoring using AI, diabetes risk prediction, and hypertension prediction.

A. Hypertension Prediction Machine Learning

Using lifestyle and demographic data, machine learning techniques are routinely utilized to predict the risk of hypertension. Age, BMI, physical activity, and smoking habits are among the variables that algorithms like Random Forest, Logistic Regression, and XGBoost examine. However, a lot of models rely on clinical records from hospitals, which restricts the general public's access to them.

B. Diabetes Prediction With AI

Using lifestyle and demographic data, including food, exercise, BMI, and family history, artificial intelligence is frequently used in early diabetes risk prediction. To facilitate early prevention, these models categorize people into risk groups. Nevertheless, a lot of current systems continue to rely on intricate clinical data.

C. Food Recognition AI- Based

Preventing diabetes and hypertension is mostly dependent on diet. AI-based food recognition systems provide automated dietary monitoring by identifying food items from photos and estimating their nutritional values using computer vision techniques like Convolutional Neural Networks (CNNs).

D. Domain-specific AI agent for Stress Monitoring

In digital health applications, conversational AI systems have grown in popularity, especially for mental health support. Natural language processing (NLP)-powered chatbots enable users to communicate with systems in natural language, providing ongoing emotional monitoring and health advice. Conversational agents are used by several digital mental health services to track emotional wellness and offer behavioral advice. To identify emotional markers like tension, anxiety, and mood swings, these systems examine user inputs. Conversational agents create tailored answers, coping mechanisms, and lifestyle suggestions targeted at enhancing mental wellness based on these insights.

The accessibility of conversational AI systems is one of their main benefits. Without having to speak with medical professionals directly, chatbot-based services enable people to get mental health support whenever they need it. Additionally, conversational systems' anonymity promotes users' more candid expression of emotional concerns, increasing participation in mental health monitoring.

Additionally, recent studies have looked into combining conversational AI with sophisticated language models and knowledge retrieval methods. Before producing responses, conversational bots can obtain domain-specific data from knowledge bases using techniques like Retrieval-Augmented Generation (RAG). This increases response accuracy and makes it possible for the system to offer more trustworthy health suggestions.

However, most existing conversational health systems provide general mental health support rather than focusing on specific lifestyle-related diseases. Conversational AI agents that are domain-specific are therefore required to assess stress levels and provide tailored advice for ailments like hypertension.

E. Connection between Stress and Hypertension

It is commonly known that stress plays a significant role in cardiovascular disorders and hypertension. Prolonged stress sets off physiological reactions that worsen long-term health and raise blood pressure. Prolonged exposure to stress can raise the risk of hypertension considerably over time.

People's stress levels are frequently measured using validated psychological evaluation instruments like the Perceived Stress Scale (PSS-10). Continuous monitoring of stress patterns can be made possible by integrating such established evaluation techniques with AI-based conversational systems. Digital health platforms can help identify lifestyle concerns early and encourage preventative health practices by combining stress detection with tailored suggestions.

III. METHODOLOGY

To monitor disease risk prediction, the suggested system gathers a variety of user data. While lifestyle-related data, such as physical activity levels and sleep habits, are retrieved through user inputs, demographic data, such as age and gender, is gathered upon user registration. User-uploaded food photos are used to collect dietary data, and computer vision algorithms are used to assess nutritional values.

Additionally, emotional indications linked to stress and well-being are captured through verbal engagements with the AI bot. In order to assess stress levels, the system also uses recognized stress assessment methods, including the Perceived Stress Scale (PSS-10). In order to produce individualized health insights and preventive suggestions, these gathered inputs are processed by machine learning and natural language processing algorithms.

A. Dataset Collection

To train and assess the machine learning models, the suggested method makes use of both user-generated and publicly accessible datasets. A publicly accessible dataset with characteristics like age, body mass index (BMI), smoking behaviors, levels of physical activity, and food patterns was utilized to predict hypertension. A dataset comprising glucose readings, BMI, family history, and lifestyle-related characteristics was used to predict diabetes.

Additionally, the convolutional neural network (CNN) model for food recognition tasks was trained using the Food-101 dataset. Accurate food item classification is made possible by the dataset's 101 food categories and numerous tagged photos. To improve personalization, user-specific data was gathered via the mobile application, including demographic data, lifestyle inputs, and conversational exchanges. To assess model performance, the gathered dataset was split into 20% for testing and 80% for training.

B. Metric Evaluation

Standard classification criteria, such as accuracy, precision, recall, and F1-score, were used to assess the machine learning models' performance. While precision and recall assess the relevance and completeness of predictions, accuracy gauges the model's overall soundness. The F1-score is appropriate for datasets that are unbalanced because it strikes a compromise between precision and recall.

Grid search approaches were used to optimize the hyperparameters of the models, which were built in Python using the Scikit-learn and XGBoost packages.

C. System Architecture

A mobile application, backend services, AI modules, and a cloud database are all integrated into the suggested platform's multi-layer design. The user interface for entering health information, uploading pictures of meals, and interacting with the conversational AI agent is provided by the mobile application layer. Data processing, machine learning model integration, and API connection are all handled by the backend layer.

A domain-specific intelligent AI agent for stress monitoring, computer vision models for food recognition, and machine learning models for diabetes and hypertension risk prediction are just a few of the analytical components included in the AI processing layer. Using sentiment analysis and natural language processing, the AI bot examines conversational answers to identify stress-related emotional cues. The user's stress level is ascertained by combining these insights with formal stress assessments.

The database layer keeps track of user health information, past interaction data, and a well-curated knowledge base that the conversational agent uses to make suggestions based on solid evidence.

D. Domain Specific AI Agent

The essential element of the suggested system is the intelligent AI agent that is domain-specific. Through conversational contact, the agent intends to track the user's stress levels and offer tailored suggestions for managing stress and preventing hypertension.

The following is a description of the AI agent's internal operation. Natural Language Processing (NLP) methods, such as tokenization and text embedding, are used in the implementation of the conversational AI agent to extract significant features from user inputs. Sentiment analysis is used to categorize user answers into positive, neutral, and negative emotional tones. Retrieval-Augmented Generation (RAG), which uses a Large Language Model (LLM) to generate responses after pertinent domain-specific knowledge is obtained from a curated knowledge base, improves response accuracy. Sentiment scores and answers from the Perceived Stress Scale (PSS-10) are used to determine the stress level. Users are categorized into low, moderate, and high stress levels based on this combined analysis, allowing for tailored recommendations.

The AI agent divides users into groups like low stress, moderate stress, and high stress based on the detected stress level. After that, tailored advice is given, including ideas for physical activity, relaxation methods, and behavioral counseling targeted at lowering stress and enhancing general health.

Through frequent user contacts, the conversational AI bot facilitates ongoing stress monitoring in addition to making recommendations. The system may detect shifts in stress levels and offer prompt treatments by examining past conversation patterns and user reactions over time. Through individualized advice, this continuous monitoring strategy encourages users to adopt healthier lifestyle practices and allows for the early detection of potential risk factors for hypertension.

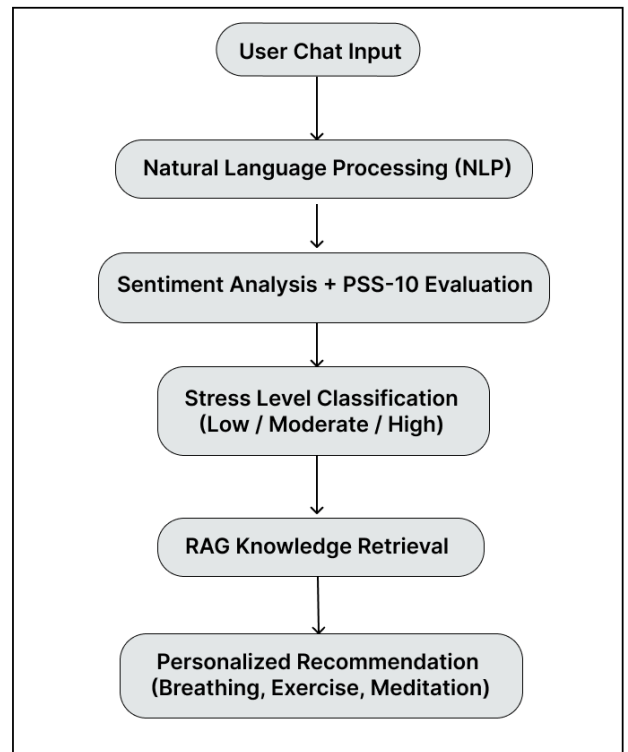


Figure 1: AI Agent Workflow

Figure 1 illustrates an AI agent analyzing user messages and recognizing emotional indicators like tension, anxiety, or mood swings using Natural Language Processing (NLP) approaches. While standardized tests like the PSS-10 questionnaire are used to gauge subjective stress levels, sentiment analysis is utilized to determine the emotional tone of user responses.

E. Usage of Technology

Table I. Proposed System Technology

Component	Technology	Purpose
Mobile App	React Native	User Interface
Backend	Python/FastAPI	API communication
Machine Learning	Random Forest, Logistic Regression	Risk prediction for Hypertension and diabetes
Computer Vision Model	Convolutional Neural Network (CNN)	Nutrient and food recognition
Conversational AI Agent	RAG + Large Language Model	Monitor stress and personalized recommendations
Database	Firebase	Store user data

Modern web and artificial intelligence technologies are used in the system's implementation to guarantee efficiency and scalability. React Native and the Expo framework were

used in the development of the mobile application to enable cross-platform interoperability for iOS and Android devices. Python and FastAPI are used to construct backend services, which make it easier to communicate with APIs and integrate them with machine learning models.

Convolutional Neural Networks (CNNs) for food recognition, machine learning techniques for disease risk prediction, and a Large Language Model with Retrieval-Augmented Generation to power the conversational AI agent are some of the artificial intelligence components. A Firebase cloud database, which offers safe and scalable data management, houses user information and system resources.

IV. Proposed System

The suggested system is a mobile health platform powered by artificial intelligence that aims to assist Sri Lankans in preventing diabetes and hypertension at an early stage. The system monitors lifestyle factors, analyzes health-related data, and offers tailored recommendations by integrating many AI technologies into mobile applications. The platform creates a comprehensive preventative healthcare solution by integrating conversational AI, computer vision techniques, and predictive analytics.

Four core functional components make up the system architecture: a domain-specific intelligent AI agent for stress monitoring, AI-based food recognition with nutritional analysis, diabetes risk prediction, and hypertension risk prediction. Together, these elements evaluate user health data and produce suggestions meant to lower the risk of chronic illnesses.

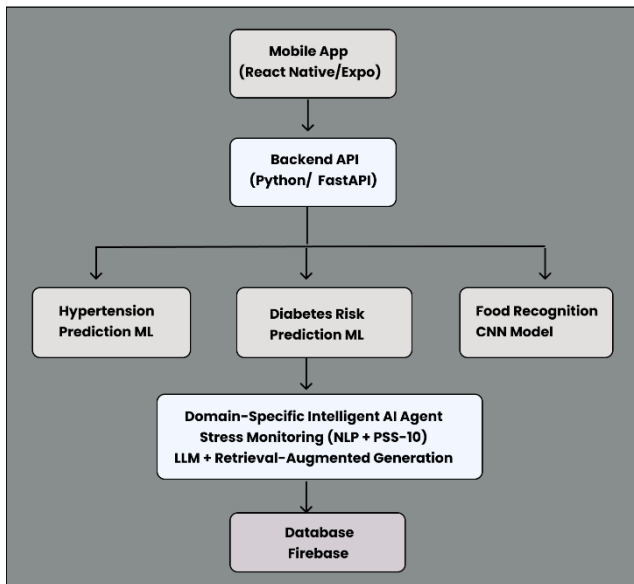


Figure 2: Proposed AI-driven Mobile App

1. Hypertension Risk Predict

The hypertension risk prediction module uses machine learning algorithms to assess the risk of acquiring hypertension by analyzing user demographic and lifestyle data, including age, gender, BMI, physical activity, and

lifestyle habits. Users are divided into three risk categories (low, moderate, and high) based on the analysis. In order to lower the risk of hypertension, the system then offers tailored advice on things like stress reduction, dietary changes, and physical activity.

2. Diabetes Risk Predict

The diabetes risk prediction module uses machine learning models to assess the risk of acquiring diabetes by analyzing lifestyle factors like food, physical activity, BMI, and family medical history. Users are categorized into various risk levels and given early warnings based on the study. To help with early illness prevention, the system also offers preventive recommendations, such as dietary adjustments, increased physical activity, and lifestyle alterations.

3. Food Recognition System

This module supports nutritional monitoring for the prevention of diabetes and hypertension by analyzing user-uploaded meal photos. The system recognizes food items and calculates their nutritional values, including calories, carbs, sodium, and fats, using computer vision and Convolutional Neural Networks (CNNs). The method offers tailored nutritional advice based on the analysis to encourage better eating practices and lower the risk of chronic illness.

4. Domain-specific AI Agent for Stress Monitoring

The primary contribution of the suggested system is a domain-specific intelligent AI agent for stress monitoring. Through conversational interaction, this component is intended to continuously monitor the user's stress levels and offer tailored suggestions for managing stress and preventing hypertension. Through a chatbot interface built into the mobile application, the AI agent communicates with users. Natural language communication between users and the system enables the agent to collect data regarding lifestyle conditions and emotional health. The device can gather contextual data that could point to health hazards associated with stress, thanks to verbal contact.

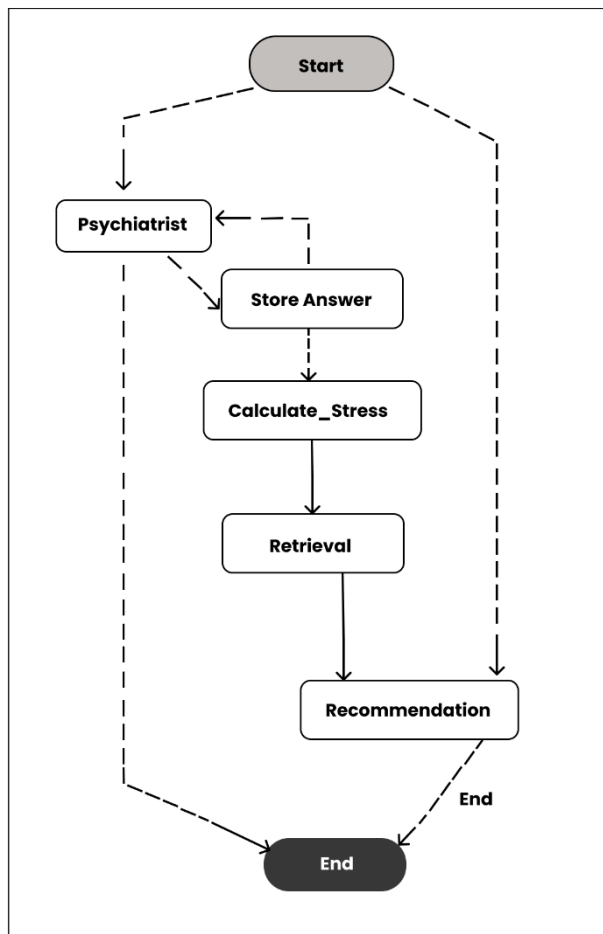


Figure 3: AI Agent Architecture Diagram

The architecture of the suggested AI-powered mental health evaluation and recommendation system is shown in Figure 3. An AI agent created to operate as a virtual psychiatrist initiates the procedure by conversing with users about ten predetermined psychological subjects. The system gathers user replies in a natural and interactive way by using this dialogue-based method. A specific stress evaluation module uses these reactions, which are methodically kept, to determine the user's stress level. A Retrieval-Augmented Generation (RAG) process is used to extract contextually appropriate recommendations from a vector database after stress computation. In parallel, the system retrieves user-specific data from a Firebase database, including hobbies and personal information. To produce tailored and context-aware recommendations, a recommendation module then combines insights from the RAG-based retrieval and user profile data. Ultimately, the system presents these customized recommendations to the user, making sure that personal preferences and psychological evaluation are considered.

The AI agent uses Natural Language Processing (NLP) techniques to evaluate human answers and identify emotional signs like tension, worry, or negative attitude. The emotional tone of user conversations is assessed via sentiment analysis. The method measures perceived stress levels using proven psychological assessment instruments like the Perceived Stress Scale (PSS-10) in addition to conversational analysis.

To produce dependable answers, the AI agent combines a Large Language Model (LLM) with Retrieval-Augmented Generation (RAG). The system uses this architecture to extract domain-specific data from a curated knowledge base that includes medical recommendations for stress reduction, preventing hypertension, and leading a healthy lifestyle. After that, the retrieved data is utilized to create customized responses that give users evidence-based advice.

The method divides users into three stress categories—low stress, moderate stress, and severe stress—based on the identified stress indicators. Every area has specific suggestions meant to lower stress levels and encourage better habits. These suggestions could include lifestyle guidance, relaxation methods, physical activity recommendations, and breathing exercises. The AI agent's capacity to facilitate ongoing stress monitoring is another crucial characteristic. The system may identify trends and shifts in emotional well-being by examining user replies and conversation history across time. This enables the platform to offer prompt treatments in the event that elevated stress levels are detected. Continuous monitoring encourages users to adopt preventative health behaviors and aids in the early identification of potential risk factors for hypertension.

The suggested AI agent offers a clever and user-friendly way to monitor stress and encourage preventative healthcare by fusing conversational AI, verified stress assessment techniques, and domain-specific health information.

V. Result and Evaluation

A. Hypertension Prediction

Several machine learning methods, such as Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Multi-Layer Perceptron (MLP), and Random Forest classifiers, were used to assess the hypertension prediction module. With an accuracy of 87.58%, the Random Forest classifier outperformed the other models, according to the trial data. The accuracy of the SVM classifier was 84.73%, compared to 85.06% for the MLP classifier. With an accuracy of 65.77%, the KNN classifier performed relatively poorly. These findings show that complicated interactions between lifestyle factors and the risk of hypertension are better captured by ensemble-based models like Random Forest.

B. Diabetes Prediction With AI

Several machine learning models, such as XGBoost, Gradient Boosting, Random Forest, Logistic Regression, KNN, and SVM, were used to assess the diabetes prediction module. With an accuracy of 98.55%, precision of 98.57%, recall of 98.55%, and an F1-score of 98.56%, XGBoost outperformed the other models. Strong performance was also shown using Random Forest and Gradient Boosting, with accuracy values over 97%.

These findings show that sophisticated boosting algorithms, which offer dependable classification performance based on lifestyle and demographic data, are very useful for predicting diabetes risk.

The performance comparison of the various machine learning models employed in this study is compiled in Table II. According to the data, Random Forest also produced good results for predicting hypertension, but XGBoost performed the best.

Table II. Machine Learning Models

Model	Accuracy	Precision	Recall	F1-score
Random Forest	87.58%	86.90%	87.20%	87.00%
SVM	84.73%	84.10%	84.50%	84.30%
MLP	85.06%	85.00%	84.80%	84.90%
KNN	65.77%	66.10%	65.50%	65.80%
XGBoost	98.55%	98.57%	98.55%	98.56%

C. AI Food Recognition model

Convolutional Neural Network (CNN)-based picture classification methods were used to assess the AI-based food recognition module. Reliable calculations of nutritional values such as calories, carbs, and fats was made possible by the model's high accuracy in detecting food items from user-uploaded photos.

The findings demonstrate that automated dietary monitoring may be successfully supported by the food identification system, eliminating the need for manual meal logging and enhancing user convenience.

D. Domain-specific AI agent

Functional testing and expert validation were used to assess the domain-specific intelligent AI agent for stress monitoring. The system's performance was evaluated based on its capacity to identify stress levels and offer pertinent recommendations during testing with actual user interactions.

Using sentiment analysis and natural language processing (NLP) in conjunction with standardized evaluation techniques like the Perceived Stress Scale (PSS-10), the AI agent effectively examined conversational inputs. The technology produced individualized stress management advice and correctly identified user stress levels based on these inputs.

A mental health expert also assessed the system and verified that the AI agent offers pertinent and suitable solutions for stress-related illnesses. Technology can promote early stress detection and encourage behavioral changes that may lower the risk of hypertension, according to expert validation.

E. Overall System

The integrated outcomes of every module show that the suggested system can provide precise forecasts and significant health insights. A holistic approach to preventive healthcare is made possible by the integration of

conversational AI, computer vision, and machine learning models. The primary contribution of this research is the AI stress-monitoring agent, which improves the system by enabling individualized interventions and ongoing emotional monitoring, features uncommon in current healthcare systems.

VI. Discussion

The potential of combining various AI technologies to enhance preventive healthcare is demonstrated by the suggested AI-driven mobile health platform. By combining conversational AI, computer vision techniques, and machine learning models, the system provides a comprehensive approach to monitoring lifestyle factors associated with diabetes and hypertension. The capacity of the suggested platform to encourage early illness prevention by recognizing risk variables before the emergence of clinical symptoms is one of its main advantages. Users are also encouraged to embrace healthier lifestyle practices by the incorporation of tailored health suggestions.

The system's domain-specific intelligent AI agent for stress monitoring, which permits ongoing monitoring of emotional well-being through conversational engagement, is another significant benefit. The conversational AI agent enables users to interact organically with the system and receive tailored stress management advice, in contrast to conventional health monitoring programs that solely depend on questionnaires or recurring health tests. A unified system for tracking stress levels, food patterns, and health hazards is also made possible by the integration of several modules into a single platform.

Despite these benefits, there are a few drawbacks to take into account. Dataset constraints may have an impact on the predictive models' and food identification system's accuracy, especially when training data is sparse or not entirely reflective of real-world situations. Additionally, if users don't routinely interact with the system or don't supply enough data for accurate analysis, there may be issues with user engagement. Continuous model refinement and the creation of user-friendly interfaces that promote regular user participation are necessary to address these limitations.

VII. Conclusion

This study introduces an AI-powered mobile health platform that uses intelligent health monitoring and tailored advice to promote the early prevention of diabetes and hypertension. The system incorporates several artificial intelligence technologies, such as conversational AI for stress monitoring, computer vision methods for food recognition, and machine learning models for disease risk prediction.

The main contribution of the suggested system is a domain-specific intelligent AI agent for stress monitoring. The AI agent enables ongoing stress level monitoring and offers tailored advice targeted at lowering the risk of hypertension by fusing natural language processing, sentiment analysis, and approved stress assessment methods. This strategy

demonstrates conversational AI's potential as a useful tool for lifestyle management and preventive healthcare.

All things considered, the suggested platform demonstrates how artificial intelligence can promote healthier lifestyle choices and early health awareness among Sri Lankans. The system offers a useful and approachable option for proactive illness prevention by combining several health monitoring components into a single mobile application.

To assess system performance in real-world healthcare scenarios, future work will concentrate on conducting user testing. Wearable health devices might be integrated for real-time physiological monitoring, and machine learning models could be further refined to increase prediction accuracy and customization capabilities.

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